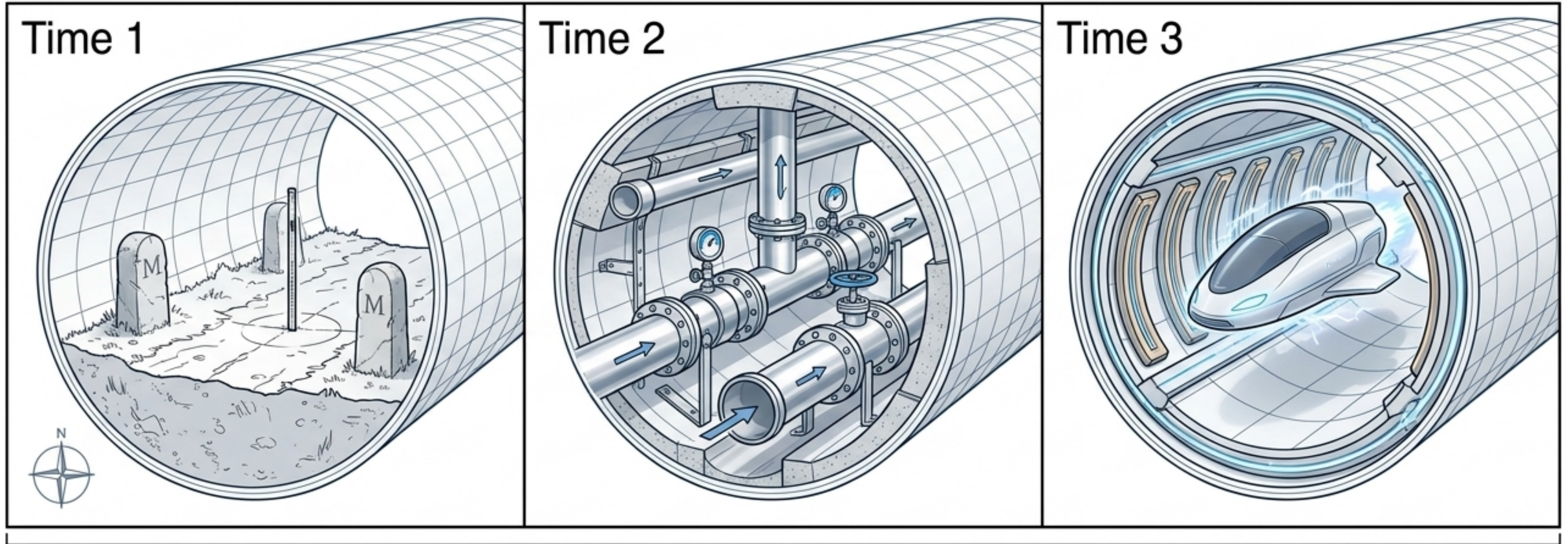


Orthodromic Infrastructure Blueprint

A Mathematical Protocol for Planetary Civil Convergence



Geometry outlasts technology

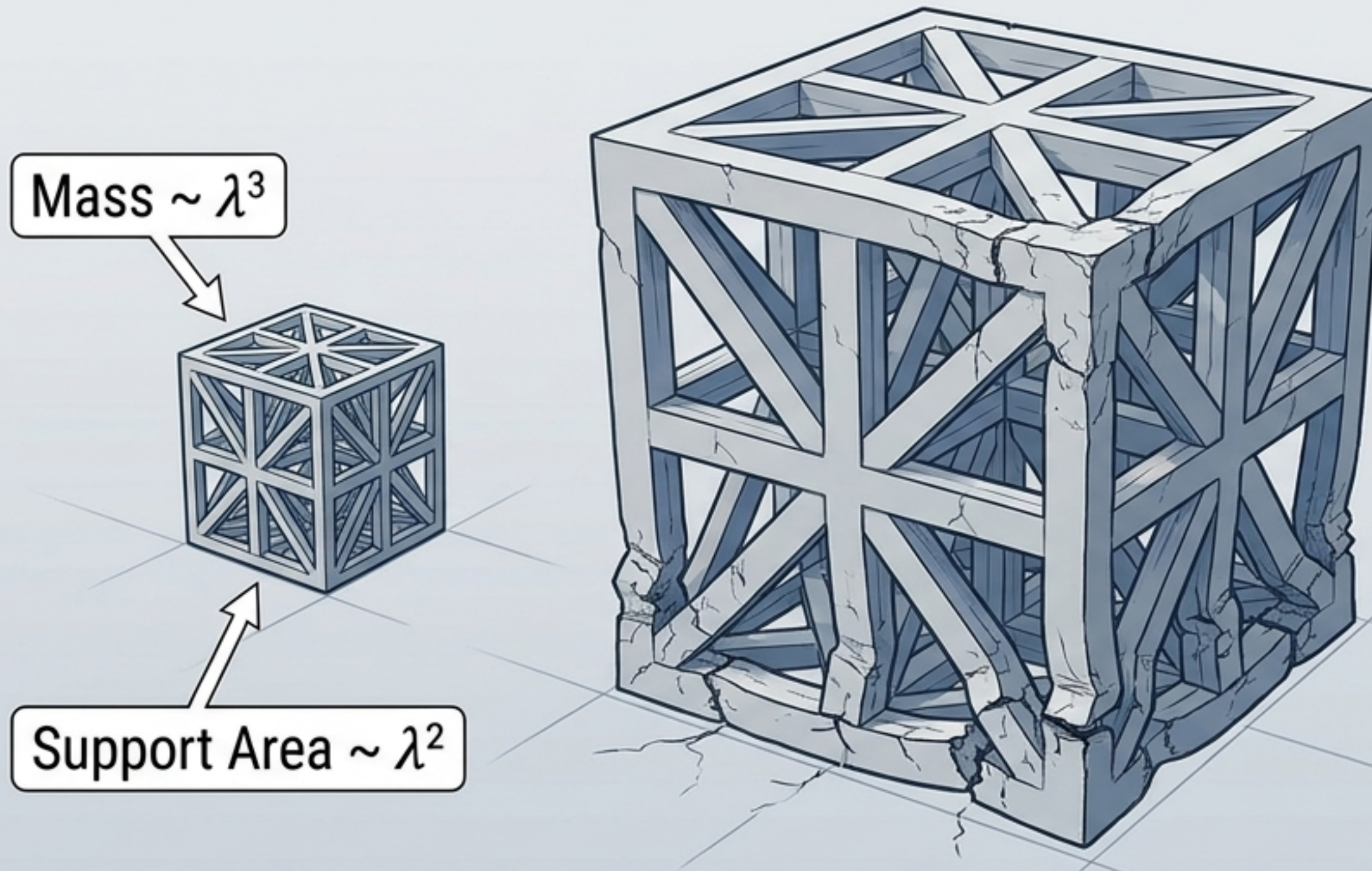


$$I_{t_{k+1}} \sim_{\Pi} I_{t_k}$$

Design Principle: Protocol Before Implementation

The blueprint shall define invariant spatial relations, interface obligations, and compatibility conditions without prematurely fixing the technologies by which those obligations are realized. A compliant implementation is identified by the preservation of required relations rather than by resemblance to a particular historical machine.

The physical limits of mechanical scaling



Derivation: Square-Cube Scaling Constraint

Suppose a structural member has characteristic length L , density ρ , and a geometrically similar cross-section. Its mass satisfies

$$m \propto \rho L^3.$$

Its weight is therefore

$$W \propto \rho g L^3.$$

If the supporting cross-sectional area satisfies

$$A \propto L^2,$$

then the compressive stress generated by self-weight obeys

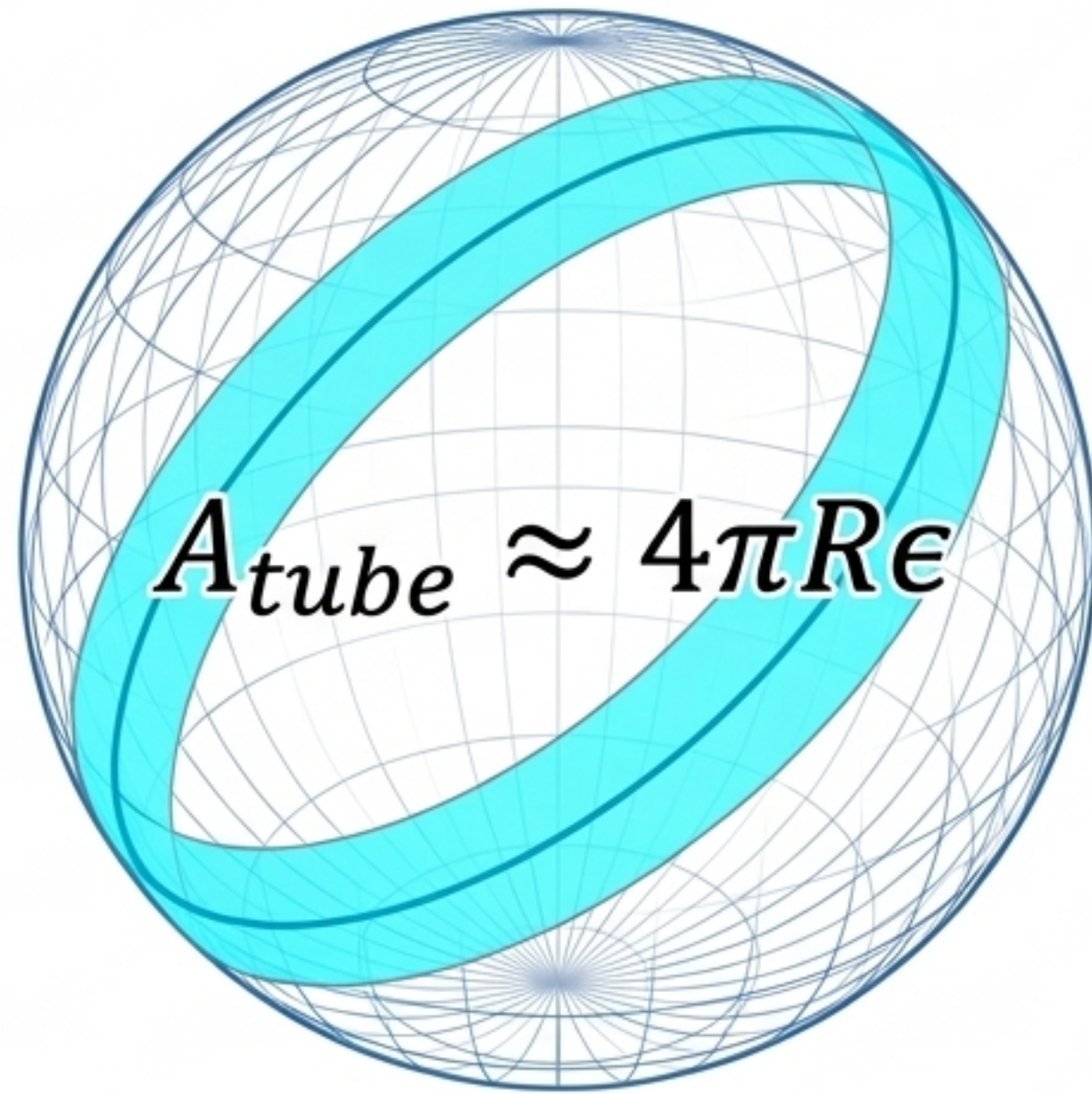
$$\sigma = \frac{W}{A} \propto \frac{\rho g L^3}{L^2} = \rho g L.$$

Thus a geometrically similar enlargement by a factor λ multiplies the self-weight stress by approximately λ . Structural similarity alone does not imply mechanical similarity.

Structural similarity alone does not imply mechanical similarity. The protocol preserves topological modularity prodularity precisely because material realization must constantly adapt to scale.

The Reference Layer vs. The Realization Layer

Reference Layer



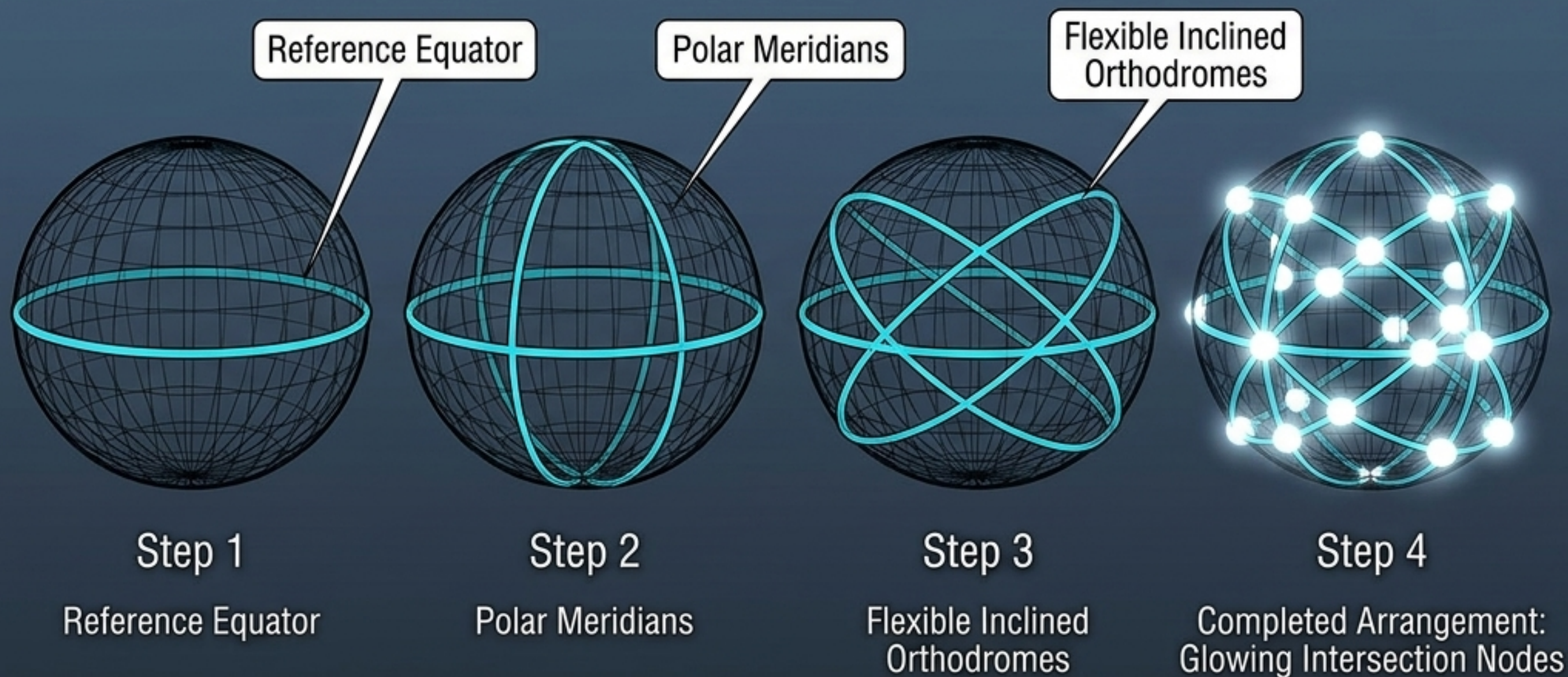
Realization Layer



Specification: Reference Alignment and Realization Deviation

Every primary corridor shall be associated with a unique reference orthodrome and a declared admissible tube $\mathcal{T}_\epsilon(O)$. A realized segment may depart from the centerline when required by terrain, geology, ecology, settlement avoidance, or structural safety, but it shall remain within its declared tube except at formally registered bypass sections. A bypass shall include entry and exit interfaces that preserve the identity and continuity of the reference corridor.

Constructing the 9-orthodrome global skeleton



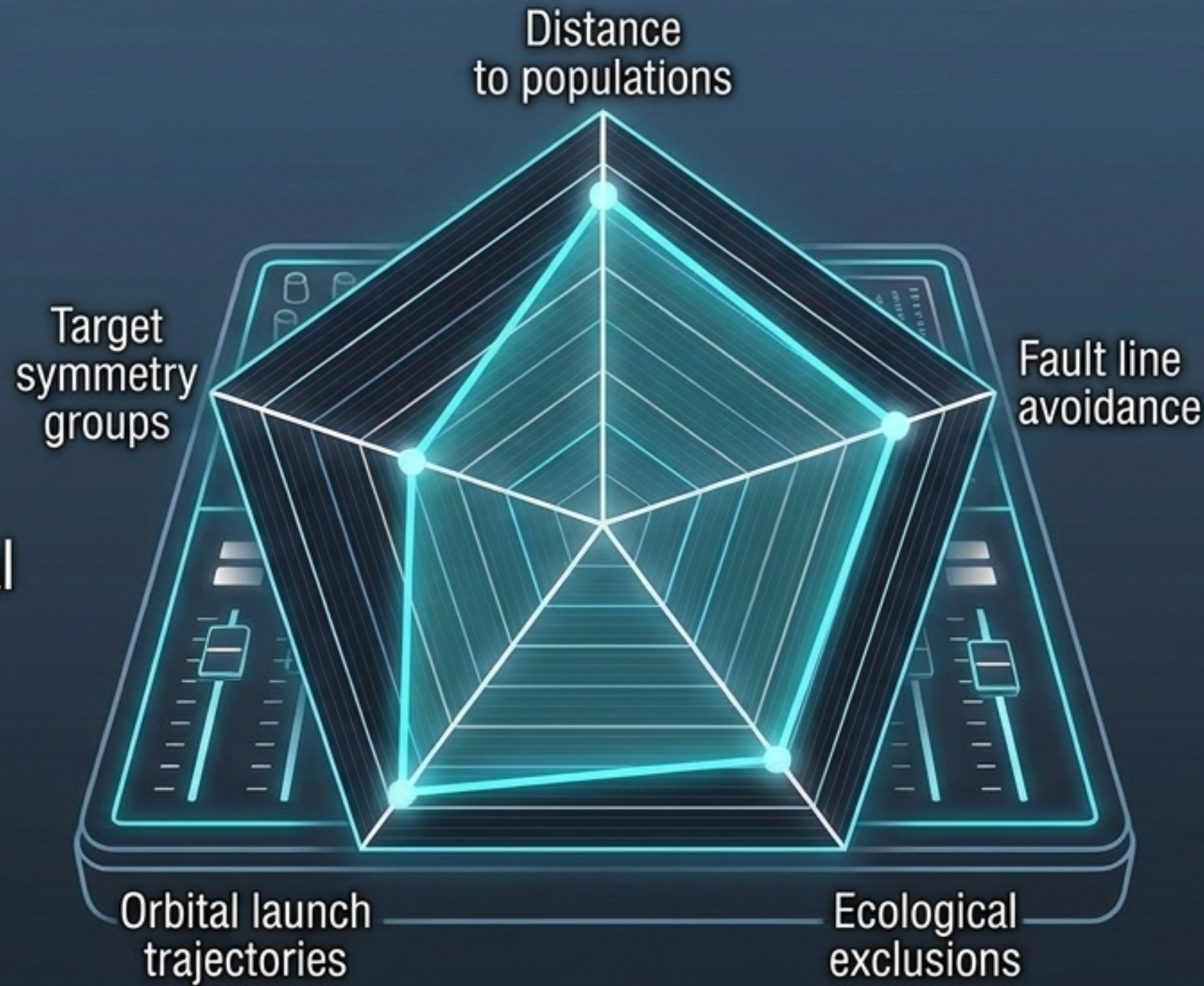
Theoretical Maxima
for $n=9$ Orthodromes

Vertices	$V = 72$
Edges	$E = 144$
Cells	$F = 74$

The protocol
preserves
topological
modularity precisely
because material
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to scale.

Optimizing the flexible geometry for the unknown future

The layout of the 4 inclined orthodromes is not arbitrary; it is the solution to a constrained variational problem across these dimensions.



Definition: Pareto Set

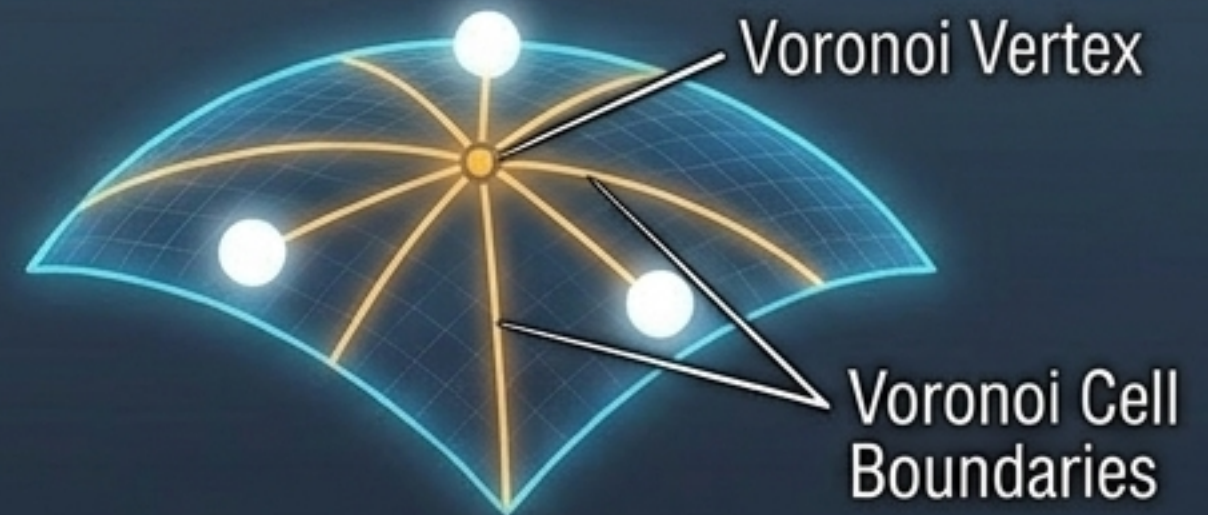
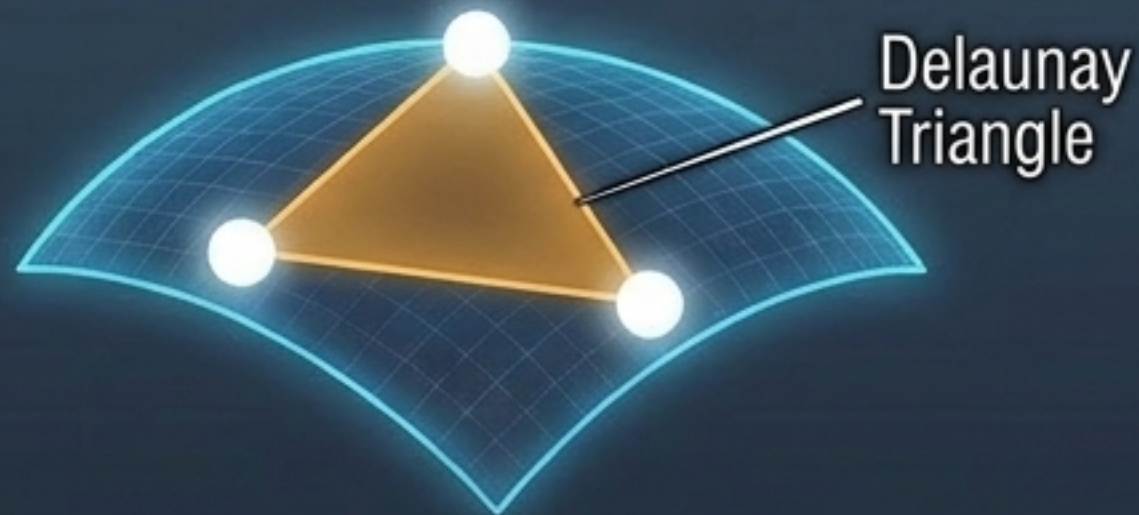
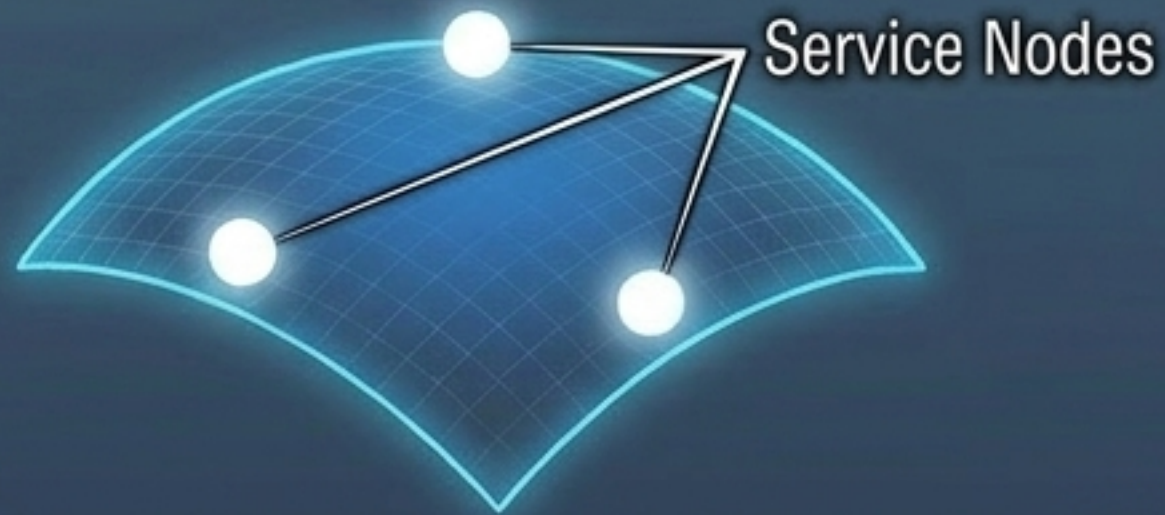
The Pareto set is

$$\mathcal{P} = \{\theta \in \Theta_{\text{adm}} \mid \exists \theta' \in \Theta_{\text{adm}} \text{ with } C(\theta') < C(\theta)\}.$$

Design Principle:
Pareto Transparency

No weighted optimum should be presented as uniquely dictated by geometry when its location depends on political or ethical weights. The protocol should preserve the Pareto frontier and disclose the weights or priority rules by which a particular configuration is selected.

Spherical service geometry: The Voronoi/Delaunay Duality



Definition: Spherical Delaunay Triangle

A triple (p_i, p_j, p_k) defines a spherical Delaunay triangle when there exists an empty spherical circumdisk whose boundary contains p_i, p_j, p_k .

The curvature defect prevents global uniformity

Derivation: Curvature Defect in a Spherical Triangulation
For each Delaunay vertex p_i , let d_i denote its degree. Since

$$\sum_{i=1}^m d_i = 2e,$$

and

$$e = 3m - 6,$$

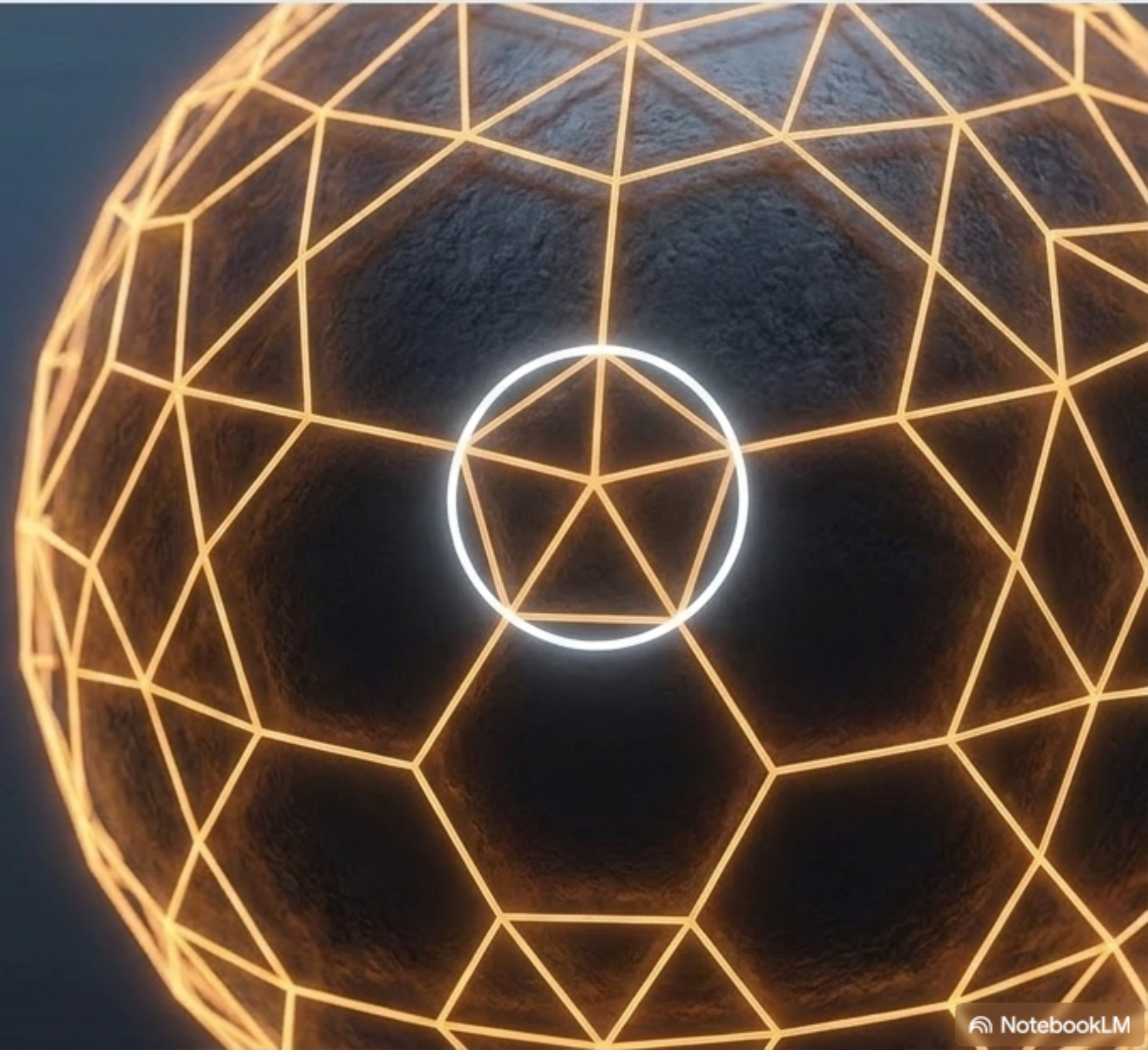
we obtain

$$\sum_{i=1}^m d_i = 6m - 12.$$

Therefore

$$\sum_{i=1}^m (6 - d_i) = 6m - (6m - 12) = 12.$$

A perfectly hexagonal triangulation is impossible on a sphere. Positive curvature dictates a total combinatorial degree defect of 12. These forced exceptional nodes represent geometric singularities or high-centrality hubs. Deviations from uniformity are mathematically inevitable.



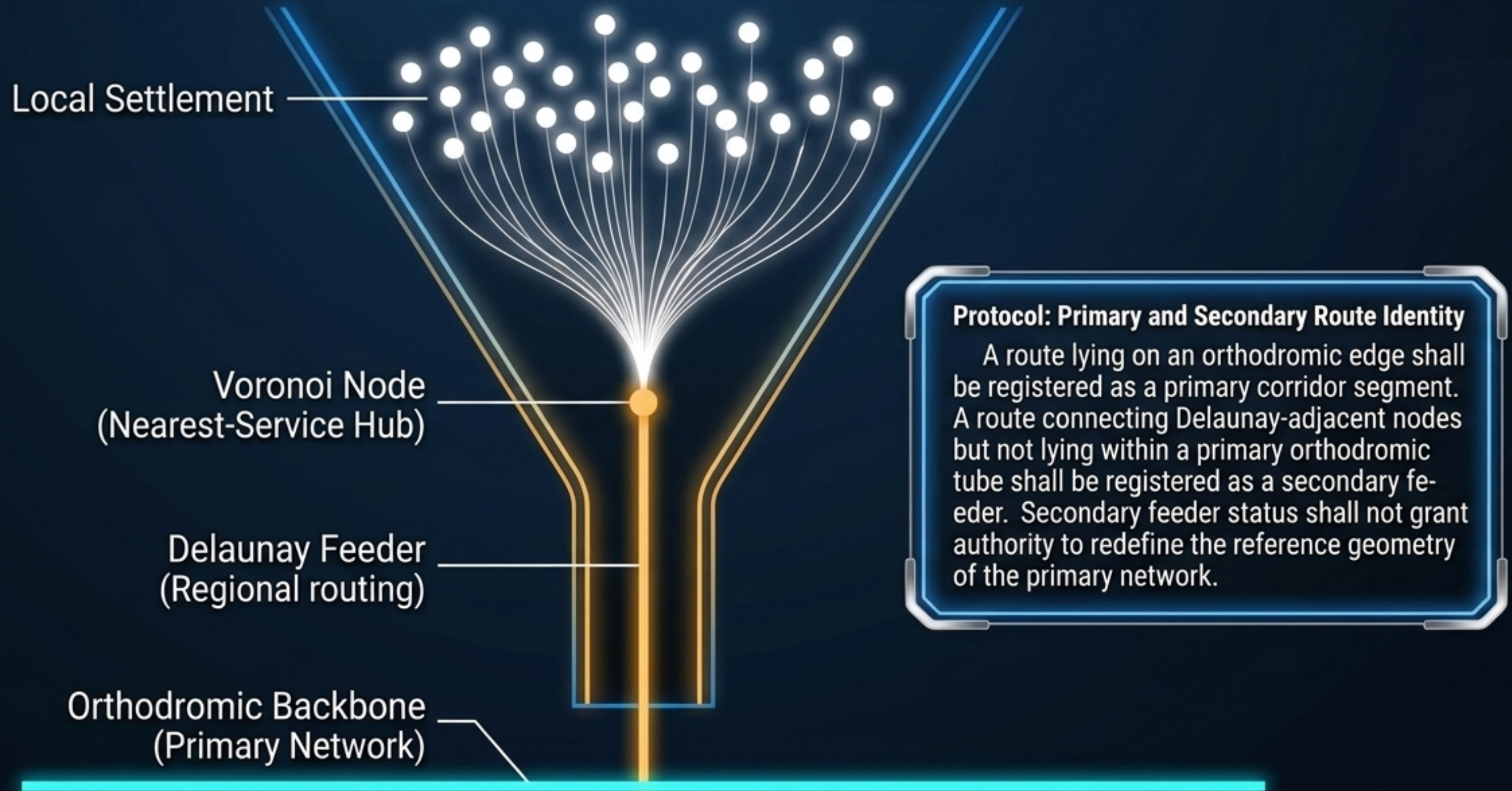
The Three-Layer Connectivity Architecture

	Basis	Function	Protocol
Primary Layer (E_O)	<u>Orthodrome</u>	Fixed Global Skeleton	Preserves global identity
Secondary Layer (E_D)	<u>Voronoi/Delaunay</u>	Adaptive Connective Tissue	Minimizes local service separation
Tertiary Layer (E_R)	<u>Terrain/Contingent</u>	Regional Adaptation	Accommodates contingent regional requirements

Design Principle: Fixed Skeleton, Adaptive Tissue

The orthodromic arrangement should act as a stable planetary skeleton, while the Delaunay and regional layers act as adaptive connective tissue. Global convergence does not require every

Routing hierarchy: From local settlement to global backbone



The Operational Microcell: Synthesizing protocol and service

Orthodromic
(C_α)

Voronoi
($\text{Vor}(p_i)$)

Intersection
($C_\alpha \cap \text{Vor}(p_i)$)

Common refinement:
 $Q_{\alpha i} = C_\alpha \cap \text{Vor}(p_i)$

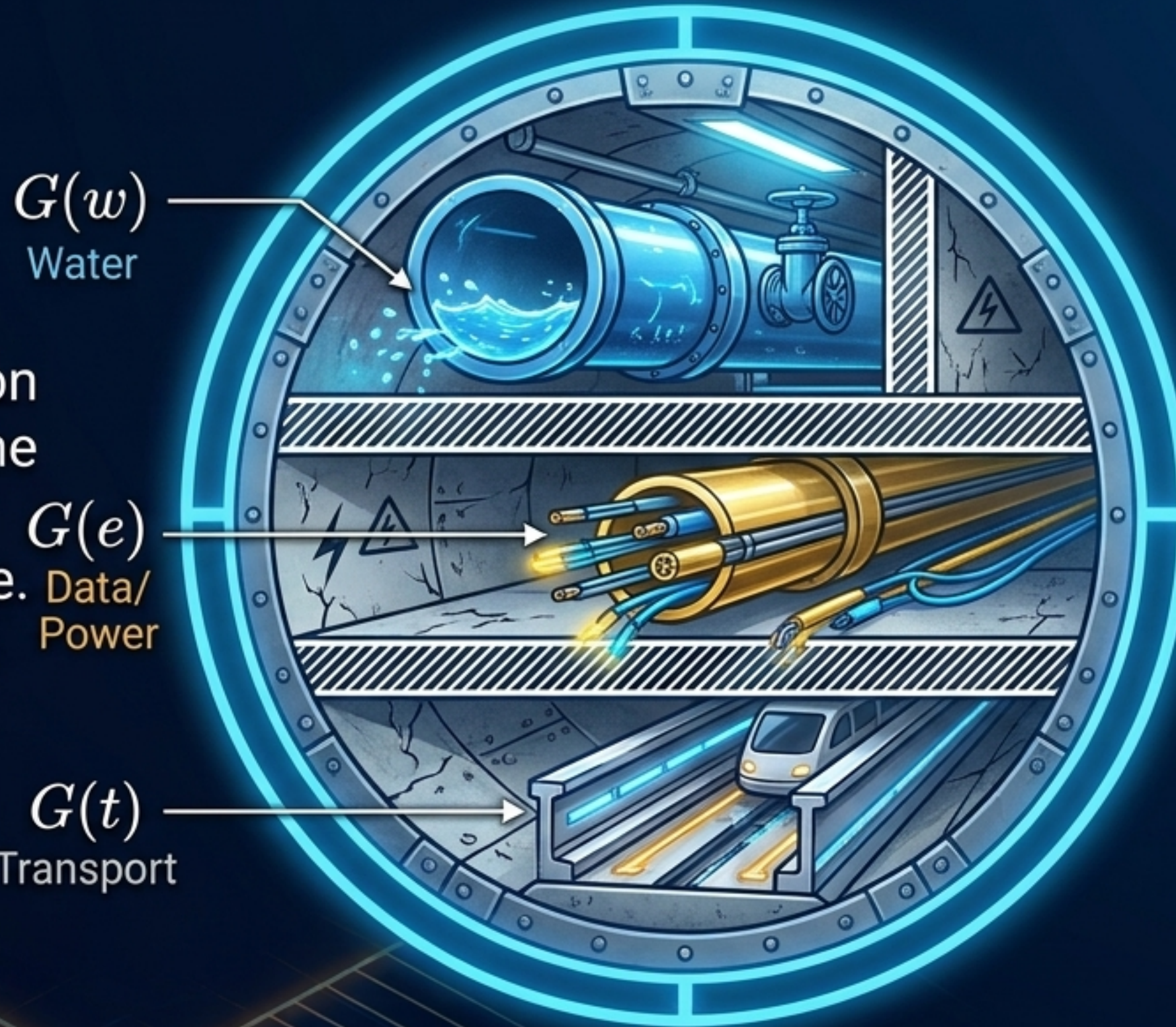
Definition: Operational Microcell

An operational microcell is a connected component of

$$C_\alpha \cap \text{Vor}(p_i)$$

Multi-commodity co-location

Spatial co-location means sharing the geometric real estate of the tube. It does not mean sharing a failure domain.



Design Principle: Shared Alignment, Separated Failure Domains

A water rupture must not automatically destroy electrical transmission. Spatial proximity shall be balanced against hard physical separation and emergency isolation.

Designing for long-horizon minimax regret



A blueprint intended to persist across centuries must prioritize robustness over nominal, short-term optimization.

A slightly inferior present-day configuration is vastly superior if its performance remains acceptable across a wide range of unknown futures.

Design Principle: Long-Horizon Minimax Regret

The placement of primary geometry should avoid decisions whose apparent present advantage depends on narrow assumptions about future population, technology, climate, or political organization. Where uncertainty cannot be reduced, the preferred configuration should minimize irreversible regret rather than maximize short-term efficiency.